VOL. I · NO. 6 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

THURSDAY, OCTOBER 29, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

One launch since Monday; Vulcan returns November 13

The only orbital launch worldwide since Monday's issue was SpaceX's Starlink 15-16, which left Vandenberg at 3:25 a.m. EDT Tuesday. That brings 2026 to 251 orbital launch attempts, 244 of them successful, according to GCAT. ISRO's GSLV with the NVS-03 navigation satellite is set for 2:21 p.m. Friday, and Starlink 15-18 follows from Vandenberg late Friday night.

United Launch Alliance says Vulcan will return to flight on Friday, November 13, carrying 40 Amazon Leo satellites, after a nine-month grounding over solid rocket booster nozzle problems. On the International Space Station, NASA's Anil Menon and the Canadian Space Agency's Joshua Kutryk began a spacewalk at 8:13 a.m. Thursday to install a new solar array.

I · RECENT LAUNCHES & RETURNS

Starlink 15-16 flies on a 26th-flight booster

Falcon 9 lifted off 2 hours 40 minutes into its window and landed its booster in the Pacific.



A Falcon 9 lifts off from Space Launch Complex 4E at Vandenberg Space Force Base with NASA's SWOT satellite on Dec. 16, 2022. Starlink 15-16 flew from the same pad on Tuesday.

PHOTO: NASA/KEEGAN BARBER · PUBLIC DOMAIN

Falcon 9: Starlink 15-16

Falcon 9 lifted off from Space Launch Complex 4E at Vandenberg Space Force Base at 3:25 a.m. EDT Tuesday (07:25 UTC; GCAT), 12:25 a.m. local time, Spaceflight Now posted. Monday's issue listed the start of the FAA airspace window, 12:45 a.m. EDT (FAA). Booster B1082 was on its 26th flight (NASASpaceflight). The booster entered service in January 2024 and has also flown the USSF-62, OneWeb Launch 20, NROL-145 and Transporter-18 missions plus 21 Starlink batches; it landed on the drone ship Of Course I Still Love You off Baja California, Mexico, and the rocket headed southeast toward an orbit inclined 70 degrees (Spaceflight Now).

"Deployment of 27 @Starlink satellites confirmed," SpaceX posted at 4:29 a.m. It was SpaceX's 122nd launch of the year (GCAT).

Elsewhere

No other orbital launches have taken place worldwide since Saturday night's Long March 8A from Wenchang, which Monday's issue covered (Launch Library 2; GCAT). Rocket Lab said its 98th Electron, for HawkEye 360 last Friday, will be followed by its 99th launch next week (Rocket Lab).

A GSLV and a Starlink on Friday, seven more by Thursday

Every orbital launch with a date between now and November 30. Times are EDT through November 1 and EST after, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Fri Oct 30, 2:21 p.m. 18:21 UTC	GSLV Mk II	Indian Space Research Organisation (ISRO)	GSLV-F18 with NVS-03, a second-generation NavIC navigation satellite	Second Launch Pad, Sriharikota, India	GTO	CONFIRMED	ISRO; Launch Library 2
Fri Oct 30, 11:48 p.m. 03:48 UTC Oct 31	Falcon 9	SpaceX	Starlink 15-18: 27 Starlink satellites	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Launch Library 2
Mon Nov 2, 3:45 a.m. EST 08:45 UTC	Electron	Rocket Lab	"The Sea God Surveys": iQPS radar imaging satellite	Launch Complex 1, Mahia, New Zealand	LEO	TENTATIVE	Launch Library 2
Mon Nov 2, 7:10 p.m. EST 00:10 UTC Nov 3	Kinetica 2	CAS Space	Listed as a Spacesail (G60) polar group; payload not confirmed	Launch Area 140, Jiuquan, China	Polar	TENTATIVE	Launch Library 2
Mon Nov 2, 8:42 p.m. EST 01:42 UTC Nov 3	Falcon 9	SpaceX	USSF-432, a classified U.S. Space Force mission (FAA: "R-4")	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Launch Library 2
Wed Nov 4, 12:00 a.m. EST 05:00 UTC	Small launcher (name provisional)	Agency for Defense Development, South Korea	Demonstration flight, first full orbital launch of the military small-satellite rocket	Offshore launch platform	LEO	TENTATIVE	Launch Library 2
Wed Nov 4, 4:00 p.m. EST 21:00 UTC	Ceres-2	Galactic Energy	Payload not announced	Launch Area 95A, Jiuquan, China	—	TENTATIVE	Launch Library 2
Thu Nov 5, 3:00 p.m. EST 20:00 UTC	Long March 12	CASC	Nine SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	TENTATIVE	Launch Library 2
Thu Nov 5, 11:59 p.m. EST 04:59 UTC Nov 6	Falcon 9	SpaceX	Starlink 15-21: 27 Starlink satellites. Launch Library 2 lists 9:21 p.m.	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	FAA; Launch Library 2
Mon Nov 9, 10:03 p.m. EST 03:03 UTC Nov 10	Falcon 9	SpaceX	"R-5," a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	Launch Library 2
NET Tue Nov 10	Electron	Rocket Lab	"The Mountain God Rises": iQPS radar imaging satellite	Launch Complex 1, Mahia, New Zealand	LEO	PRELIMINARY	Launch Library 2
Fri Nov 13	Vulcan VC6L	United Launch Alliance	Vulcan Leo 1 (LV-01): 40 Amazon Leo broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	CONFIRMED	ULA; Amazon Leo
NET Mon Nov 23	Ariane 64	Arianespace	VA271: 36 Amazon Leo broadband satellites, 10th Ariane 6	ELA-4, Kourou, French Guiana	LEO	PRELIMINARY	Launch Library 2
Tue Nov 24, 12:03 a.m. EST 05:03 UTC	Soyuz-2.1b	Roscosmos	Progress MS-36 cargo ship to the International Space Station	Site 31/6, Baikonur, Kazakhstan	LEO	TENTATIVE	Launch Library 2
NET Wed Nov 25	Long March 2F	CASC	Shenzhou 24 crew to the Tiangong station	Launch Area 91, Jiuquan, China	LEO	PRELIMINARY	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; CASC, China Aerospace Science and Technology Corp.; EST, Eastern Standard Time, which begins November 1.

Date changes

Starlink 15-18. Friday, October 30, at 12:02 a.m.; now Friday at 11:48 p.m., with a backup window starting at 11:33 p.m. Saturday (FAA). No reason given.

Starlink 15-21. Thursday, November 5, at 12:14 a.m. EST; now late Thursday, with an FAA airspace window opening at 11:59 p.m. EST (FAA). No reason given.

NROL-195. Thursday, November 5, at 5:34 p.m. EST; now listed only for November (Launch Library 2). No reason given.

Electron "The Sea God Surveys." NET November 3; now Monday, November 2, at 3:45 a.m. EST (Launch Library 2).

Vulcan with Amazon Leo. November, no date; now Friday, November 13, "pending range approval" (ULA).

Long March 12A Flight 2. NET November 3 in Monday's issue; no longer in the upcoming schedule of Launch Library 2.

New to the schedule. USSF-432 and "R-5" from Vandenberg, Kinetica 2, Ceres-2, the Long March 12 and the South Korean demonstration flight, which now has a date (Launch Library 2).

Also listed for November without a specific date in Launch Library 2: Starship Flight 15 from Starbase; NROL-195, OneWeb 21 and Northrop Grumman's Cygnus NG-25 on Falcon 9s; and ISRO's first uncrewed Gaganyaan test flight on LVM-3 and a PSLV-XL with the TDS-01 technology demonstration satellite.

QUICK HIT

Starship. "Ship 42 has completed both its single- and six-engine static fire tests, paving the way for Flight 15 in November," NASASpaceflight posted Tuesday evening.



Launch sites with orbital launches listed through November 15; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary in Launch Library 2; for Cape Canaveral's Vulcan, ULA has since announced November 13. Vandenberg also launched since Monday.
MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Airspace windows for three Vandenberg Falcons

The FAA's operations plan lists the times U.S. airspace is set aside; it gives no hazard-area coordinates.

How they fly

Tuesday's Falcon 9 headed southeast from California's central coast toward an orbit inclined 70 degrees, and its booster landed on a drone ship off Baja California (Spaceflight Now). The three Falcon 9s in the FAA's plan all launch from the same pad, Space Launch Complex 4E. Friday's GSLV is bound for geostationary transfer orbit from India's Second Launch Pad at Sriharikota (Launch Library 2).

Airspace windows

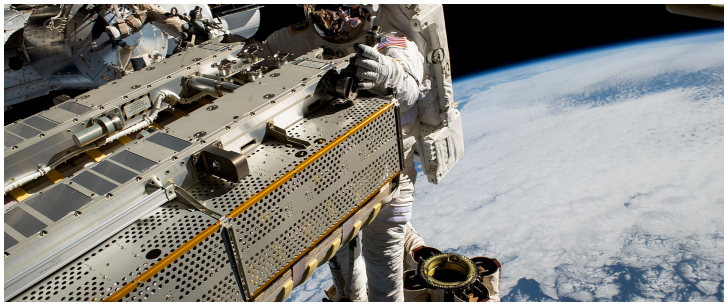
The FAA Air Traffic Control System Command Center's plan for Thursday lists these launch windows, converted from UTC (FAA advisory):

FLIGHT	PRIMARY WINDOW	BACKUP
Starlink 15-18	**Fri Oct 30, 11:48 p.m.–Sat 4:31 a.m.**	**Sat Oct 31, 11:33 p.m.–Sun Nov 1, 3:16 a.m. EST**
"R-4" (USSF-432)	Mon Nov 2, 8:42 p.m.–Tue 1:25 a.m. EST	Tue Nov 3, 8:28 p.m.–Wed 1:11 a.m. EST
Starlink 15-21	Thu Nov 5, 11:59 p.m.–Fri 4:42 a.m. EST	Fri Nov 6, 9:06 p.m.–Sat 1:49 a.m. EST

A 1973 French satellite comes down after 52 years

CelesTrak lists Aureole 2, five Starlinks and four other objects as decayed since Monday's issue.

Station traffic: a seventh roll-out array



NASA astronaut Woody Hoburg works on a Roll-Out Solar Array outside the International Space Station during a 2023 spacewalk.
PHOTO: NASA · PUBLIC DOMAIN

"The spacewalk is now underway!" NASA posted at 8:13 a.m. Thursday. NASA astronaut Anil Menon and Canadian Space Agency astronaut Joshua Kutryk are to install and unfurl the seventh of eight International Space Station Roll-Out Solar Arrays, or IROSAs (Spaceflight Now), in a spacewalk planned to last about six and a half hours (NASA). Spacewalk preparations and robotics work filled Tuesday for the Expedition 75 crew (NASA).

The next visitors are unchanged: Progress MS-36 at 12:03 a.m. EST November 24 from Baikonur, Cygnus NG-25 in November and China's Shenzhou 24 to Tiangong NET November 25 (Launch Library 2).

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Monday's issue. Dates are UTC; no reentry-time estimates were available for this issue.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
Rassvet-3 23 (100089)	Jul 19, 2026	Oct 24	291 × 300
Starlink-30224 (57416)	Jul 20, 2023	Oct 24	185 × 196
Long March 6A debris (54346)	Nov 11, 2022	Oct 24	246 × 270
Object D, 2018-102D (43834)	Dec 7, 2018	Oct 25	158 × 171
Starlink-2187 (48562)	May 15, 2021	Oct 25	158 × 168
Starlink-36255 (67102)	Dec 15, 2025	Oct 25	205 × 210
Starlink-2236 (48582)	May 15, 2021	Oct 26	167 × 176
Aureole 2 (7003)	**Dec 26, 1973**	**Oct 27**	**147 × 159**
Starlink-1036 (44741)	Nov 11, 2019	Oct 27	160 × 169
Iperdrone.0 (60520)	Aug 16, 2024	Oct 27	167 × 179

Aureole 2, a French-owned satellite launched from Plesetsk, came down on October 27, the day Monday's issue listed as its predicted reentry (CelesTrak). Rassvet-3 23, which Monday's issue listed as predicted for November 6, is recorded as having decayed on October 24, before that issue appeared (CelesTrak). Iperdrone.0 is Italian-owned and was launched from Vandenberg; Object D, an unidentified object from a 2018 launch at Jiuquan (CelesTrak).

Still up

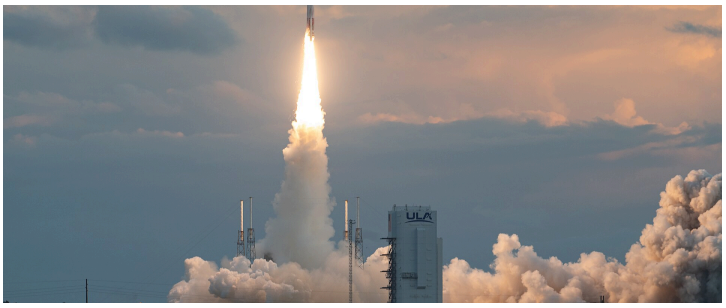
Rassvet-3 17, from the same July 19 launch and listed Monday with a predicted reentry of 12:16 a.m. Friday, is still in orbit, at 187 by 192 kilometers and 82.3 degrees inclination (CelesTrak).

Vulcan returns to flight with six boosters and fixed nozzles

ULA's November 13 flight carries Amazon's largest Leo payload yet, and Rocket Lab's Neutron hardware reached its Virginia pad.

United Launch Alliance

The launch of a Vulcan VC6L "carrying the next batch of production satellites for @AmazonLeo, Vulcan Leo 1, is planned for Friday, Nov. 13, pending range approval," ULA posted Wednesday. "All 40 Leo satellites—our largest payload to date—have been integrated with Vulcan inside Amazon's vertical integration facility (VIF-A) at the Cape," Amazon Leo posted.



Vulcan lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station on its second certification flight on Oct. 4, 2024.
U.S. SPACE FORCE PHOTO BY AIRMAN 1ST CLASS COLLIN WESSON · PUBLIC DOMAIN

It will be Vulcan's fifth flight and its first with six strap-on solid rocket boosters, after a nine-month grounding. Nozzle failures on the Northrop Grumman-built boosters marred its second flight and, in February, its fourth, though both reached their intended orbits, Spaceflight Now reported. "This upcoming launch will utilize the improved nozzle design, and additional testing and analysis will continue in parallel with the mission," ULA said in a statement quoted by Spaceflight Now. Amazon has booked 38 Vulcan launches, nearly 400 Leo satellites are in orbit, and a second vehicle is already assembled for Vulcan Leo 2 (Spaceflight Now).

Rocket Lab

"Neutron's Stage 2 and Thrust Module have arrived to Virginia," Rocket Lab posted Wednesday; after a barge ride from its Neutron factory they were offloaded on Wallops Island, with integrated fluids testing and launch-mount checkouts next. The company said its 100th Electron is "scheduled for delivery to Launch Complex 1 soon" (post), that eight of 18 Tranche 2 spacecraft for the Space Development Agency are in build in California (post), and that its solar panels on NASA's NEO Surveyor sunshade deliver up to 2 kilowatts, the spacecraft's only power source (post).

Starship

Ship 43 has returned to the production site from chopstick tests at Pad 2, and Booster 23 was set to head to Massey's for cryogenic testing; the pair is expected to sail to Kennedy Space Center for the first Starship flight from Pad 39A, Flight 16, NASASpaceflight reported.

QUICK HIT

Artemis III. "The core stage of the Artemis III rocket is officially stacked," NASA's Artemis account posted Wednesday.

Europe and India

ESA's Hera "can now see Didymos and Dimorphos as two separate asteroids," the agency posted Tuesday; the first detailed images of Dimorphos are expected on November 25. ESA has ordered four more Galileo Second Generation satellites, bringing the total to 16 (ESA), and signed three contracts in Norway on GNSS interference testing, maritime navigation and drones (post). ISRO set GSLV-F18 for 23:51 India time Friday, 2:21 p.m. EDT (post).

Comment deadlines

New Glenn at SLC-36. Comments on raising New Glenn's cap from 12 to 50 launches a year close Friday, October 30 (notice, docket FAA-2026-9841).

Part 450. Comments on five proposed changes to the commercial launch and reentry licensing rule are due November 4 (docket FAA-2026-9935).

FCC space licensing. Comments on the Space Modernization for the 21st Century further notice (SB Docket No. 25-306) are due November 25 and replies December 28 (Federal Register).

Starbase. Scoping comments on up to 456 Starship launches a year at Starbase are due November 30, with meetings November 17 to 19 (Federal Register).

VI · BY THE NUMBERS

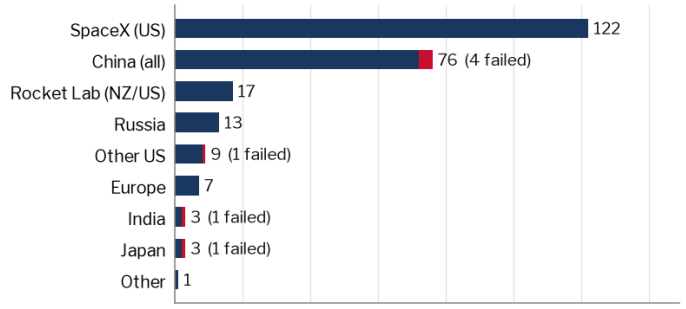
251 orbital launch attempts so far in 2026

All 19 orbital launches in October have succeeded, and 10 of them were SpaceX Falcons.

251	244	122	76	19
orbital launch attempts worldwide, Jan 1–Oct 27	reached orbit	by SpaceX, all successful	by China, four failures	orbital launches in October, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 251 orbital launch attempts through Tuesday's Starlink 15-16, with 244 successes. SpaceX has flown 122, all successful, followed by China with 76, Rocket Lab with 17 and Russia with 13; Europe has flown seven, other U.S. providers nine, and India and Japan three each.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 27, 07:25 UTC

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. Galactic Energy's Ceres-2 is listed to fly again on November 4 (Launch Library 2).

October so far

Ten of this month's 19 orbital launches were SpaceX Falcons (GCAT). Vandenberg has hosted six of them, more than any other site, and Cape Canaveral and Kennedy four between them. China has flown six, while South Korea's Nuri, Japan's H3 and Rocket Lab's Electron account for the other three.

DATE, EDT	ROCKET	MISSION	SITE
Oct 1, 11:10 a.m.	Falcon 9	Crew-13	Cape Canaveral
Oct 1, 2:32 p.m.	Falcon 9	Transporter-18 rideshare	Vandenberg
Oct 1, 11:54 p.m.	Falcon Heavy	NROL-97	Kennedy
Oct 6, 11:25 p.m.	Nuri	NeonSat-2 to 6	Naro, South Korea
Oct 9, 3:27 p.m.	Long March 12	SatNet LEO Group 27	Wenchang, China
Oct 10, 3:39 a.m.	Falcon 9	SDA Tranche 1 Transport Layer A	Vandenberg
Oct 12, 8:28 p.m.	Falcon 9	Starlink 15-25	Vandenberg
Oct 13, 6:33 a.m.	Falcon 9	CRS-35	Cape Canaveral
Oct 14, 7:40 a.m.	Zhuque-2E	10 satellites	Jiuquan, China
Oct 15, 8:01 p.m.	Falcon 9	USSF-481	Vandenberg
Oct 19, 3:41 p.m.	H3-24	MMX	Tanegashima, Japan
Oct 20, 7:04 a.m.	Falcon 9	Bandwagon-5 rideshare	Cape Canaveral
Oct 20, 11:39 p.m.	Ceres-1S	Xiguang-4 01	Sea, off Rizhao, China
Oct 23, 3:11 a.m.	Electron	HawkEye 360 (3 satellites)	Mahia, New Zealand
Oct 23, 6:15 a.m.	Long March 6A	Yaogan-50 03	Taiyuan, China
Oct 23, 3:25 p.m.	Long March 4B	Six Yaogan satellites	Jiuquan, China
Oct 23, 6:48 p.m.	Falcon 9	Starlink 15-15	Vandenberg
Oct 24, 10:03 p.m.	Long March 8A	SatNet LEO Group 28	Wenchang, China
Oct 27, 3:25 a.m.	**Falcon 9**	**Starlink 15-16**	**Vandenberg**

The week ahead

Nine orbital launches have specific dates between Friday and November 5 (Launch Library 2; FAA): three SpaceX Falcon 9s from Vandenberg, three Chinese rockets, ISRO's GSLV, Rocket Lab's 99th Electron and South Korea's offshore demonstration flight. If Friday's GSLV and Starlink 15-18 fly on schedule, October would end with 21 orbital launches and SpaceX would reach 123 for the year.

COMING UP

Fri., Oct. 30 · 2:21pm EDT GSLV with NVS-03, Sriharikota (ISRO)

Fri., Oct. 30 · 11:48pm EDT Starlink 15-18, Vandenberg (FAA)

Mon., Nov. 2 · 8:42pm EST USSF-432, Vandenberg (FAA)

Fri., Nov. 13 Vulcan Leo 1, Cape Canaveral (ULA)

Takeaways

1. Starlink 15-16 was the only orbital launch since Monday, for 251 attempts this year.
2. Vulcan returns to flight November 13 with 40 Amazon Leo satellites and redesigned booster nozzles.
3. ISRO's GSLV and Starlink 15-18 both fly Friday.
4. An ISS spacewalk Thursday is installing the seventh of eight roll-out solar arrays.
5. Aureole 2, launched in 1973, reentered on October 27.

KEY DATES AHEAD (EDT, THEN EST FROM NOV 1)

Fri Oct 30	GSLV with NVS-03, 2:21 p.m.; Starlink 15-18, 11:48 p.m.; New Glenn comments close
Sun Nov 1	Eastern Standard Time begins
Mon Nov 2	Electron iQPS, 3:45 a.m.; Kinetica 2, 7:10 p.m.; USSF-432, 8:42 p.m.
Wed Nov 4	South Korean demo flight; Ceres-2; Part 450 comments due
Thu Nov 5	Long March 12, 3 p.m.; Starlink 15-21, 11:59 p.m.
Fri Nov 13	Vulcan Leo 1
Wed Nov 25	Hera's first detailed Dimorphos images expected; FCC comments due

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, ESA, ISRO, FAA, FCC, Federal Register. Tracking: CelesTrak, GCAT (Jonathan McDowell). X posts: @SpaceflightNow, @NASASpaceflight, @RocketLab, @SpaceX, @NASA, @Space_Station, @NASAArtemis, @esa, @isro, @ulalaunch, @AmazonLeo. Photos: NASA and U.S. Space Force (public domain). Maps: Natural Earth (public domain).

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